Pretrained Models We train six size variants of BLOOM with respective hyperparameters detailed in Table 3. Architecture and training hyperparameters come from our experimental results (Le Scao et al., 2022) and prior work on training large language models (Brown et al., 2020; Kaplan et al., 2020). Model depth and width for the non-176B models roughly follow previous literature (Brown et al., 2020; Zhang et al., 2022), deviating for 3B and 7.1B in order only to fit the models more easily on our training setup. Embedding parameter sizes are larger for BLOOM owing to the larger multilingual vocabulary, but scaling literature discounts embedding operations (Kaplan et al., 2020). During the development process at the 104B parameters scale, we experimented with different values of Adam β parameters, weight decay and gradient clipping to target stability, but did not find it helpful. For all models, we use a cosine learning rate decay schedule (Loshchilov and Hutter, 2016) over 410B tokens, taken as an upper bound for the length of training if compute permitted, and warmup for 375M tokens. We use weight decay, gradient clipping, and no dropout. The ROOTS dataset contains around 341 billion tokens of text, so we aimed to train all models for the equivalent amount of tokens. However, in light of revised scaling laws published during training (Hoffmann et al., 2022), we decided to train the large models for an additional 25 billion tokens on repeated data. As warmup tokens + decay tokens were larger than the total number of tokens, the end of learning rate decay was never reached.

Multitask Finetuning Finetuned BLOOMZ models (Muennighoff et al., 2022b) maintain the same architecture hyperparameters as BLOOM models. The finetuning hyperparameters are loosely based on T0 (Sanh et al., 2022) and FLAN (Wei et al., 2021). Learning rates are determined by doubling the minimum learning rate of the respective pretrained model and then rounding. Global batch sizes are multiplied by four for small variants to increase throughput. While the models are finetuned for 13 billion tokens, the best checkpoint is chosen according to a separate validation set. We found performance to plateau after $1 - 6$ billion tokens of finetuning.

Contrastive Finetuning We also perform contrastive finetuning of the 1.3 and 7.1 billion parameter BLOOM models using the SGPT Bi-Encoder recipe (Muennighoff, 2022) to train models that produce high-quality text embeddings. We created SGPT-BLOOM-7.1B-msmarco[24] geared towards multilingual information retrieval and SGPT-BLOOM-1.7B-nli[25] for multilingual semantic textual similarity (STS). However, recent benchmarking has found these models to also generalize to various other embedding tasks, such as bitext mining, reranking or feature extraction for downstream classification (Muennighoff et al., 2022a).

3.5.1 Carbon Footprint

While most attempts to estimate the carbon footprint of language models have shed light on the emissions produced due to energy consumed during model training (e.g. Patterson et al., 2021; Strubell et al., 2019), other sources of emissions are also important to consider. In our efforts to estimate the carbon emissions of BLOOM, we were inspired by the Life Cycle Assessment (LCA) approach (Klöpffer, 1997) and aimed to consider aspects such as

24. `hf.co/bigscience/sgpt-bloom-7b1-msmarco`
25. `hf.co/bigscience-data/sgpt-bloom-1b7-nli`

the emissions of equipment manufacturing, intermediate model training, and deployment. According to our estimates, the carbon emissions from BLOOM training add up to approximately 81 tons of CO_2eq, of which 14% were generated by the equipment manufacturing process (11 tons), 30% by the energy consumed during training (25 tons) and 55% by idle consumption of the equipment and computing cluster used for training (45 tons).

Model name	**Number of parameters**	**Power consumption**	**CO_2eq emissions**
GPT-3	175B	1,287 MWh	*502 tons*
Gopher	280B	*1,066 MWh*	*352 tons*
OPT	175B	*324 MWh*	70 tons
BLOOM	176B	433 MWh	25 tons

Table 4: Comparison of carbon emissions between BLOOM and similar LLMs. Numbers in *italics* have been inferred based on data provided in the papers describing the models.

Comparing the carbon emissions of BLOOM training to other similar models (see Table 4), reveals that while the energy consumption of BLOOM is slightly higher than OPT (Zhang et al., 2022) (433 Mwh compared to OPT's 324 MWh), its emissions are approximately 2/3 less (25 tons versus 70 tons). This is thanks to the low carbon intensity of the energy grid used for training BLOOM, which emits 57 gCO_2eq/kWh, compared to 231 gCO_2eq/kWh for the grid used for OPT training. Specifically, France's national energy grid (which is used by Jean Zay) is largely powered by nuclear energy, which is low-carbon compared to grids powered by energy sources such as coal and natural gas. While the sustainability of nuclear energy is debated, it is one of the least carbon-intensive sources of energy that is currently available. Both BLOOM and OPT incurred significantly less carbon emissions than GPT-3 (as reported by (Patterson et al., 2021)), which can be attributed to several factors including more efficient hardware as well as less carbon-intensive energy sources.

We also pursued further exploration of the carbon footprint of (1) the computation carried out on Jean Zay within the scope of the Big Science workshop, and (2) running the BLOOM model API in real time. In terms of the footprint of the totality of the computation, we estimate that the final BLOOM training represents approximately 37% of the overall emissions, with other processes such as intermediate training runs and model evaluation adding up to the other 63%. This is slightly less than the estimate made by the authors of the OPT paper, who stated that the total carbon footprint of their model is roughly 2 times higher due to experimentation, baselines and ablation (Zhang et al., 2022). Our ongoing exploration of the carbon emissions of the BLOOM API have estimated that the real-time deployment of the model on a GCP instance with 16 GPUs running in the `us-central1` region results in approximately 20 kg of CO_2eq emitted per day of deployment (or 0.83 kg per hour). This figure is not representative of all deployment use-cases, and will vary depending on the hardware used as well as the specifics of model implementation (e.g. whether batching is used) and the number of requests the model receives. Further information regarding BLOOM's carbon footprint can be found in Luccioni et al. (2022).

3.6 Release

Openness has been central to the development of BLOOM and we wanted to ensure it is easily available for the community to use. As such, we worked on producing documentation as a Model Card (Mitchell et al., 2019) and a new license addressing specific goals of the project.

Model Card Following best practices for releasing machine learning models, the BLOOM model has been released along with a detailed Model Card[26] (Mitchell et al., 2019) describing its technical specifications, details on training, intended-use, out-of-scope uses as well as the model's limitations. Participants across working groups worked together to produce the final Model Card and similar cards for each checkpoint. The work was collaborative, primarily composed "live" by thinking through and discussing each section, then further dividing into subsections based on the categorizations and distinctions participants naturally ended up creating throughout discussions.

Licensing Being mindful of the potentially harmful use-cases that BLOOM could enable, we chose to strike a balance between unrestricted open-access and responsible-use by including behavioral-use clauses (Contractor et al., 2022) to limit the application of the model towards potentially harmful use-cases. Such clauses are routinely being included in a growing class of "Responsible AI Licenses (RAIL)"[27] that the community has been adopting when releasing their models.[28] A distinguishing aspect of the RAIL license developed for BLOOM is that it separates licensing of the "source code" and "model", as referenced by its trained parameters. It further includes detailed definitions of "use" and "derived works" of the model to ensure that anticipated downstream use by prompting, finetuning, distillation, use of logits and probability distributions are explicitly identified. The license contains 13 behavioral-use restrictions that have been identified based on the intended uses and limitations described in the BLOOM Model Card, as well as the BigScience ethical charter. The license offers the model at no charge and users are free to use the model as long as they comply with the terms (including usage restrictions). The source code for BLOOM has been made available under an Apache 2.0 open source license.

4. Evaluation

Our evaluations focus on zero-shot and few-shot settings. Our goal is to present an accurate picture of how BLOOM compares to existing LLMs in settings that most realistically reflect the way the models are likely to be used in practice. Because of the scale of these models, prompt-based adaptation and few-shot "in-context learning" are currently more common than finetuning. Thus, we report results on a range of tasks - SuperGLUE 4.2, machine translation 4.3, summarization 4.4 - and languages in zero-shot and one-shot prompt-based settings, as well as after multitask finetuning (Section 4.7). We also perform code generation 4.5, use BLOOM-derived text embeddings for representation tasks 4.8 and interpret BLOOM's generalization abilities from the perspective of multilingual probing (Section 4.9).

26. `hf.co/bigscience/bloom`
27. `licenses.ai`
28. `the-turing-way.netlify.app/reproducible-research/licensing/licensing-ml.html`